

Husnain Ahmed

Passport: BP1184091 | **Date of birth:** 09/01/2003 | **Place of birth:** wah |

Nationality: Pakistani | **Phone number:** (+92) 3225290604 (Mobile) | **Phone number:** (+92) 3454785536 (Mobile) | **Email address:** husnain7810@gmail.com | **LinkedIn:** https://www.linkedin.com/in/husnain-ahmed2003/ |

Address: Qaziabad street no 04 Po Box Losar sherfo near kohistan enclave , 47000, wah, Pakistan (Home)

WORK EXPERIENCE

NETWORK ENGINEER – NAYATEL – 16/07/2026 – Current – ISLAMABAD, PAKISTAN

- Monitoring and optimizing IP/MPLS, routing, switching, and VoIP systems to ensure high availability and QoS for 200,000+ subscribers.
- Train and onboard new engineering batches/hires, building team capability alongside operational duties.
- Coordinate cross-functional capacity planning, service deployment, and incident resolution, improving SLA compliance.

OPERATION ENGINEER – NAYATEL – 06/2025 – 16/07/2026 – ISLAMABAD, PAKISTAN

• Monitored and optimized core network operations including IP/MPLS, routing, switching, and VoIP systems, ensuring high availability and reliability of services.
• Implemented and maintained network performance monitoring tools, proactively identifying and resolving faults to minimize downtime and improve QoS (Quality of Service).
• Coordinated with cross-functional teams for capacity planning, service deployment, and incident resolution, streamlining operations and improving SLA compliance impacting over 20,000+ Families.

NETWORK SUPPORT ENGINEER – NAYATEL – 08/2024 – 06/2025 – ISLAMABAD, PAKISTAN

• Provided L1/L2 technical support by diagnosing and resolving customer connectivity issues in FTTH, routing, switching, and VoIP networks, ensuring high service uptime.
• Configured, monitored, and troubleshot network devices (ONTs, routers, and switches) using tools such as ping, traceroute, and port diagnostics to maintain network performance.
• Escalated and documented complex incidents while collaborating with NOC and engineering teams, improving mean-time-to-resolution (MTTR) and customer satisfaction,Impacting Over 5000+ Families.

INTERN – ISLAMABAD ELECTRIC SUPPLY COMPANY LIMITED 7/2023 – 9/2023

- Assisted in preventive maintenance at IESCO, reducing unplanned power outages by 15%.
- Enhanced system reliability by identifying and resolving electrical faults, ensuring minimal downtime.
- Analyzed and resolved power quality issues in a cross-functional team, addressing voltage fluctuations.
- Implemented preventive maintenance schedules, decreasing equipment failures by 25%.
- Actively participated in training on grid modernization and emerging technologies.(LINK)

EDUCATION AND TRAINING

B.SC ELECTRICAL ENGINEERING University Of Engineering & Technology Taxila 2020–2024

Courses:

- Control System (Grade A)
- Industrial Automation (Grade A)
- Power Generation (Grade A)
- Power Distribution & Utilization (Grade A)
- Electric Power Transmission (Grade A)
- Power System Protection (Grade A)

- Power System Analysis (Grade A)
- High Voltage Engineering (Grade A)

Final Grade**: 3.59/4.00** | Percentage**: 89.75** (LINK)

FINAL YEAR DESIGN PROJECT CONTROL AND DESIGN OF BALL BALANCING SYSTEM 2023-2024

Problem Statement:
- Developed a Proportional-Integral-Derivative (PID) controller to stabilize a ball on a plate using a resistive touch panel for accurate position sensing.
- Addressed the challenge of maintaining equilibrium of a dynamic object through real-time control and constant adjustments.

Uniqueness:
- Integrated mechanical engineering, electrical control systems, and advanced algorithms, providing a hands-on learning experience.
- Bridged theoretical and practical realms of control systems engineering, with applications in aerodynamics and other industrial fields.

Outcomes:
- Enhanced educational understanding of control systems through real-time interaction.
- Developed a robust PID controller model applicable to various industries such as aerospace, automotive, and manufacturing.
- Contributed to sustainable development goals by promoting technological advancements and optimizing control systems for efficiency.
- Demonstrated the effectiveness of the system in achieving and sustaining balance under various conditions, validating its potential for real-world applications. (LINK)

2018 – 2020

F.SC PRE-ENGINEERING F.G SCIENCE DEGREE COLLEGE FOR MEN,WAH CANTT

Final Grade**: A** | Percentage**: 83.36 | 917/1100** (LINK)

MATRICULATION (SCIENCE) F.G PUBLIC SCHOOL NO 03 FOR BOYS,WAH CANTT

Final Grade**: A** | Percentage**: 85.27 | 938/1100** (LINK)

SKILLS

Matlab/Simulik | PVWATTS | Sketchup3d | Helioscope | Ltspice | Proteus | MPLAB | NI-MultiSim | AUTOCAD - 2D + 3D | Arduino IDE | Modelsim | PSSE | DEV C++ | TURBO C++ | Adobe Lightroom

VOLUNTEERING

15/10/2020 – 24/04/2024 UET TAXILA

SOCIETY OF INNOVATIVE ELECTRICAL PROFESSIONALS (SIEP)

Vice President (April 2023 - April 2024)
- Oversaw the organization and execution of numerous events, competitions, and seminars, enriching the learning experience for 200+ members.
- Spearheaded initiatives like the: Cricket Kit Design Competition, Electrical Engineering Online Quiz, Scholarship Opportunities Webinar, Industry Expectations for Engineers Webinar, Electronics Workshop (including soldering), Career Counselling Session at Sir Syed School, 2023 Orientation, Digital Dastak - Digital Marketing Webinar, and SIEP Mania, fostering a dynamic and engaging environment.Impacting overall 2500+ Students. (LINK)

Assistant Director, Media Club (Jan 2022 - April 2022)
- Headed media coverage for events like the 2022 Session Orientation and Sports Week.
- Served as the primary photographer, videographer, photo editor, and video editor.
- Produced the majority of the society's social media content, including videos and photos of key events, showcasing the society's activities to a wider audience.
- Initiated and organized a Photography and Videography Competition at the university, promoting creative expression.
- Delivered a Workshop on MATLAB, sharing technical knowledge with fellow students.(LINK)

Executive Member (Oct 2021 - Dec 2022)
- Managed numerous events, fostering a vibrant community within the society.
- Led the organization of games during Sports Week 2021, collaborating effectively with my team.(LINK)

General Member (Oct 2020 - Oct 2021)

- Supported team leaders in the effective management of various events, facilitating participation for over 1,500 students.
- Contributed to successful events like the 2021 Orientation Session, Speak Up Session, Robotics Session, and Online Earning Workshop.(LINK)

HONOURS AND AWARDS

14/10/2023

Recipient of Prime Minister Laptop Scheme (PMYP) – Prime Minister's Youth Laptop Scheme Phase-III

Acknowledged for exemplary academic excellence, I was honored to receive laptop through the esteemed Prime Minister Laptop Scheme (PMYP). This distinguished award is bestowed upon students who showcase remarkable dedication and excellence in their academic pursuits.(LINK)

ACADEMIC PROJECTS

20/03/2022 – 27/04/2022

Control Of Robotic Arm

Designed and developed a robotic arm in the Microprocessors & Microcontrollers Lab using wood and servo motors, achieving four degrees of freedom. The arm performs simple tasks like light material handling, enhancing safety in hazardous environments. Simulated the code in Proteus and verified PWM signals with an oscilloscope. The LCD displayed motor duty cycles based on potentiometer values, ensuring precise control. This project showcased skills in robotics, microcontroller programming, and real-time system verification.(LINK)

16/03/2024 – 24/04/2024

Automation of Agricultural Irrigation System

Developed an automated agricultural irrigation system in the Industrial Automation Lab using CX-Programmer and CX-Designer. Implemented ladder logic to streamline and automate irrigation processes, enhancing efficiency and resource management.(LINK)

16/03/2024 – 24/04/2024

Phase Reversal Protection of Three-Phase Induction Motor

Implemented Phase Reversal Protection for a three-phase induction motor using a Magnetic Contactor, Phase Sequence Relay, and other essential components in the complex engineering problem of power system protection. This project aimed to safeguard the motor against incorrect phase sequence, ensuring reliable operation and preventing potential damage.(LINK)

16/03/2024 – 24/04/2024

Gold Leaf Electroscope Project

Constructed a Gold Leaf Electroscope in the High Voltage Lab using aluminum foil for the leaves, a PVC pipe for charging, and a glass structure for housing. This project demonstrated practical application of electrostatic principles. (LINK)

09/10/2023 – 22/11/2023

PVI Measurement Module and Software Implementation of Transmission Line Model using MATLAB SIMULINK

Led the development and integration of a PVI (Phase Voltage Current) Measurement Module and implemented a Transmission Line Model using MATLAB SIMULINK in Electrical Power Transmission Lab. This project focused on improving power distribution system efficiency and reliability by detecting and analyzing phase voltage imbalances, while simulating transmission line dynamics through advanced software modeling.(LINK)

09/10/2023 – 22/11/2023

Energy Storage and Solar Farm Positioning Optimization For University Of Gujrat

Led the design of an 8 MW solar installation for the University of Gujrat, optimizing energy efficiency and positioning within a specific budget. Utilized Helioscope and real irradiation data from the PV Watts website to plan the deployment of 14,538 monocrystalline solar panels, each 550 watts. This project showcased my ability to achieve sustainable energy goals while adhering to financial constraints, emphasizing practical expertise in solar system design and cost-effective energy solutions. (LINK)

15/04/2023 – 24/05/2023

Load Flow, Power Flow Analysis, and Fault Analysis in Power Systems

Created a comprehensive project in the Power System Analysis Lab focusing on Load Flow, Power Flow Analysis, and Fault Analysis using IEEE 14-bus and IEEE 5-bus systems. Employed Newton-Raphson, Fast Decoupled, and Gauss-Seidel methods in PSS/E and MATLAB for load and fault simulations, including balanced and unbalanced faults.(LINK)

15/04/2023 – 24/05/2023

Power Factor Measurement and Correction using Capacitor Bank

Designed and implemented a Capacitor Bank in the Power Distribution and Utilization Lab to improve power factor in electrical systems. Utilized various capacitors, a choke for inductive load simulation, and light bulbs for resistive load simulation. Incorporated a 5A circuit breaker for safety and an energy meter to measure voltage, current, power, and power factor. This project showcased my expertise in power system optimization and practical application of electrical components for energy efficiency.(LINK)

14/11/2022 – 19/12/2022

LEAD, LAG Compensator, and PID Controller for Antenna Azimuth Position Control System

Designed and tuned Lead and Lag compensators to improve the dynamic response and stability of the control system.Implemented a PID controller to ensure precise position control of the antenna azimuth.Conducted system modeling and simulation using MATLAB/Simulink to analyze and verify the performance of the compensators and the PID controller.(LINK)

14/11/2022 – 19/12/2023

Amplitude Modulation and Demodulation

Developed an Amplitude Modulation and Demodulation project in the Communication System Lab. Designed the circuit initially in LTspice, then implemented it in hardware. Used components including capacitors (1.25nF, 1.45nF), an inductor (1mH), a 100kohm potentiometer, and resistors (1kohm, 2.2kohm). This project demonstrated my proficiency in analog communication techniques and practical skills in circuit design and implementation.(LINK)

14/11/2023 – 19/12/2023

Temperature and Pressure Measurement Device

Developed a project for Temperature and Pressure Measurement using a BMP180 sensor, Arduino UNO, and a 16x2 LCD Display. Initiated with coding in Arduino IDE to interface and process sensor data. This project highlighted my skills in sensor integration, microcontroller programming, and real-time data visualization through LCD displays, essential for applications in environmental monitoring and data acquisition systems.(LINK)

20/03/2022 – 27/04/2022

Audio Signal Processing with MATLAB: Real-time Recording, Delay, Attenuation, Playback, and Save

Developed a project in the Signal and System Lab focusing on real-time audio signal processing using MATLAB. Implemented functionalities including real-time audio recording, delay, attenuation, playback, and saving of processed audio signals. This project demonstrated proficiency in digital signal processing techniques and MATLAB programming for audio applications, emphasizing practical skills in system analysis and real-time data manipulation.(LINK)

18/09/2021 – 22/10/2021

DESIGN OF PASSIVE BAND PASS,BAND STOP &TWIN-T NOTCH FILTER

Developed and simulated filters in LTSpice for Electrical Network Analysis, enhancing signal processing capabilities by accurately targeting specific frequency ranges.These filters play a pivotal role in signal processing and frequency control, finding applications in various domains such as communications, audio processing, and instrumentation.(LINK)

18/09/2021 – 22/10/2021

Binary Counter Using JK Flip Flop

Developed a binary counter in ModelSim using JK flip-flops as part of a Digital Logic Design Lab project. The versatile JK flip-flop, known as a universal flip-flop, was configured to function as SR, D, and T flip-flops, demonstrating its input-following character and adaptability.(LINK)

LICENSES & CERTIFICATIONS

11/03/2024

Work Smarter with Microsoft Excel

Successfully completed the Work Smarter with Microsoft Excel course by Microsoft, a comprehensive program spanning 20 hours conducted over a period of 5 weeks. Acquired advanced skills in data analysis, virtualisation, and automation using Excel, enhancing productivity and efficiency in professional workflows.(LINK)

12/03/2024

ChatGPT Prompt Engineering Mastery

Description: Completed the ChatGPT Prompt Engineering Mastery course on Udemy, gaining expertise in utilizing the ChatGPT model effectively for various natural language processing tasks, including text generation, summarization, and dialogue systems.(LINK)

COMMUNITY WORK

17/01/2020 – CURRENT

Expert Local Guide | Google Maps Contributor

- As an accomplished Local Guide at Level 5 on Google Maps, I am deeply committed to enhancing community connectivity and facilitating seamless navigation.
- Through meticulous attention to detail and a passion for exploration, I have amassed a remarkable track record of over 100 contributions, comprising insightful reviews and visually captivating photographs.
- My contributions have left a significant mark, garnering over 60,000 views on my meticulously curated photos and more than 10,000 views on my detailed reviews. These efforts have not only facilitated the journeys of countless individuals but have also served as a beacon for community engagement and exploration.(LINK)

WORKSHOP,SEMINAR & WEBINARS

10/04/2023

Matlab Workshop For Beginners

Organized and delivered a workshop on MATLAB, covering fundamental commands, functions, tools, and techniques.Provided an introduction to Simulink, emphasizing its application in engineering simulations and system modeling.(LINK)

22/03/2023

PCB Designing Workshop by SIEP

Participated in a workshop focused on PCB (Printed Circuit Board) designing.Acquired skills in schematic design, layout, and manufacturing considerations in PCB development.(LINK)

05/02/2023

Electrical Engineering Roadmap by SIEP

Delivered a comprehensive webinar to junior students.Provided guidance on achieving a high GPA.Shared strategies for earning money online during studies.Emphasized the importance of staying active and personal growth.(LINK)

28/10/2022

Importance of CCNA by Amjad Ismail

Attended a webinar highlighting the significance of CCNA certification in networking.It covers a broad range of networking topics, including routing and switching, network security, wireless networking, and more.(LINK)

08/09/2021

Graphic Designing Trends in 21st Century

Participated in a workshop exploring contemporary trends in graphic design.Explored innovative techniques and technologies shaping graphic design practices in the 21st century.(LINK)